Examiner only

12. *(a)* Some students were given a mixture made up of about 50% each of sodium chloride and sodium iodide and asked to design tests to show the presence and concentration of iodide ions in this mixture.

(i) One student decided to try electrolysis. He dissolved some of the mixture in distilled water to give a concentrated solution. He then passed electricity through the solution, using inert electrodes. The negative ions moved to the anode where they lost electrons.

State what was seen at the anode to confirm the presence of iodide ions.
Explain your answer, including a half-equation. [3]

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(ii) Another student decided to use a chemical test. She dissolved some of the mixture in distilled water and then added a little aqueous silver nitrate. After noting what was seen, she added some aqueous ammonia to the mixture and shook it.

State and explain what occurred during this test. [3]

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(iii) In a further test the concentration of the iodide ions present was found by a titration with potassium iodate(V) dissolved in a strong acid. 11.24 g of the mixture of sodium chloride and sodium iodide were dissolved in distilled water and made up to 250 cm^3. The iodide ions present in 25.0 cm^3 of this solution required 18.00 cm^3 of the potassium iodate(V) solution of concentration 0.100 mol dm^{-3}, to completely react.

Calculate the number of moles of iodide ions (present as sodium iodide, M_r 150) in 25.0 cm^3 of the solution and hence the exact percentage of sodium iodide in the mixture. [3]

(1 mol of potassium iodate(V) reacts with 2 mol of sodium iodide)

Percentage of sodium iodide = ... %

(b) Hydrogen chloride is a colourless gas that absorbs at 278 nm in the ultraviolet region of the electromagnetic spectrum.

Use this information and the data sheet to show that the energy of the H–Cl bond is 431 kJ mol^{-1}. [2]

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(c) The table shows bond energy values and absorption maxima for the hydrogen halides.

H–X	Bond energy / kJ mol^{-1}	Absorption maximum / nm
H–F	562	212
H–Cl	431	278
H–Br	366	326
H–I	299	400
H–At		

The accepted range for wavelengths for the visible region of the electromagnetic spectrum is 400-700 nm.

Suggest and explain why it is probable that H–At would be a coloured gas. [2]

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END OF PAPER